MILITARY SPECIFICATION

COMPARATOR, BENCH TYPE, VERTICAL COLUMN, ELECTRONICALLY AMPLIFIED

1. SCOPE

1.1 Scope. This specification covers an electronically amplified, vertical column, bench type comparator.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, wood, cleated plywood.
PPP-C-843	Cushioning Material Cellulosic.

Military

MIL-P-116	Preservation - Packaging, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Storage and Shipment
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with appropriate test methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute

USASI C39.1-1964	Indicating Instruments Requirements for

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(application for copies should be addressed to American Standards Association, 10 East 40th Street, New York, New York 10016).

Underwriters Laboratories (Standards for Safety)

UL 45 Portable Electric Tools

(Application for copies should be addressed to Underwriters Laboratories Inc., 207 E. Ohio St., Chicago, IL. 60611).

American Trucking Association Inc. (ATA)

National Motor Freight Classification Rules

(Application for copies should be addressed to American Trucking Association Inc., Traffic Dept., 1616 P. St., N. W. Washington, D. C. 20036).

Uniform Classification Committee (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Sta., 516 W. Jackson Blvd., Chicago, IL 60606).

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Preproduction sample is not required.

3.2 Design. The comparator shall be new and of the quality and design currently being offered by manufacturers producing equipment as described herein. Unless otherwise specified (see 6.2), the comparator shall be a self contained portable unit designed for 115 volts A.C., single phase, 60 cycle operation. Any requirement for D.C. electrical power shall be provided for in the internal electrical circuitry of the unit. When properly assembled, the comparator shall be capable of full range repetitive gaging accuracy to within the tolerance limits specified herein.

3.2.1 Interchangeability. All parts shall be manufactured to such standards as will permit ease in adjustment and replacement without modification of parts.

3.2.2 Material. All parts used in fabrication of the gage assembly shall be new and of the quality necessary to produce a gage that will meet the requirements specified herein.

3.2.3 Safety. The equipment supplied under this specification shall have adequate guards (where applicable) for preventing injury and shall be sufficiently insulated for preventing shock.

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3.3 Construction. The comparator shall be complete so that when assembled and connected to the specified source of power it shall be ready for operation. Construction shall be free from any characteristics or defects that will prevent the comparator from passing any of the examinations and tests in Section 4.

3.3.1 Casting and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the assembly.

3.3.2 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.3 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.4 Coating or painting. Corrosion resistant treatment, painting, or plating shall be in accordance with manufacturer's standard commercial practice with the exception that there shall be no cadmium plating used on any component of the assembly.

3.4 Components. The comparator shall consist essentially of the column and base with adjusting mechanism, gaging head, column saddle, gaging anvil and the electronic amplifier.

3.4.1 Column and base. The column and base shall be an integral structure of cast iron, iron alloy or steel, properly aged and treated to insure material stability for reliable repetitive amplifier read-out to within the accuracy limits specified in 4.4. The column shall be machined to insure an accurate sliding fit for the saddle and unless otherwise specified (see 6.2) shall be of sufficient height for comparative gaging operations over a range from 0 to 20 inches in the vertical direction. The column shall be fitted with a rack and pinion gear or an adjusting screw assembly and handwheel for positioning the gaging head to any location within the gaging range. The base shall be of sufficient size to provide adequate support for the column and shall have three point contact for surface plate or bench mounting. The top of the base, adjacent to the gaging head contact point, shall be accurately machined for mounting the gaging anvil. The mounting and securing device shall insure perpendicular relation between the gaging head actuating spindle and the gaging surface of the anvil.

3.4.2 Gaging head. The gaging head shall be equipped with a fine feed adjusting mechanism (.030 inch minimum range) and a contact tip retracting device. Unless otherwise specified (see 6.2), the contact tip shall be a standard replaceable diamond of 1/8 inch radius, with actuating pressure of not less than 4 ounces and not more than 8 ounces. The actuating mechanism shall be free from backlash, sticking and friction which would produce inaccurate measurements and erroneous indications for

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repetitive gaging operations. The gaging head shall include mounting features for attaching to the column saddle. The gaging contact shall slide along a vertical axis that is perpendicular to the gaging surface of the gaging anvil.

3.4.3 Saddle. The saddle shall be accurately machined for friction-free sliding fit on the column guideways and shall include wear compensating features. The gaging head mounting surface shall be machined to insure consistent perpendicular relation between the spindle contact movement and the gaging anvil. The saddle clamping device shall insure positive locking effect at any selected position on the column, within the gaging range, and shall not adversely affect the gaging accuracy. Vertical adjustment shall be controlled through a manually operated rack pinion or a handwheel operated adjusting screw. The adjusting mechanism shall function smoothly and shall not allow the saddle to drop when the clamp is disengaged.

3.4.4 Gaging anvil. The anvil shall be tool steel hardened to 62-65 Rockwell "C", artificially aged for material stability, chrome plated and lapped (both gaging surfaces) to within 0.000010 inch flat. Unless otherwise specified (see 6.2), the gaging anvil's size shall be approximately 1-1/4 inches wide by 5 inches long. The anvil shall be reversible with a plain and serrated gaging surface and fitted with an auxiliary block having three point contact. The auxiliary block shall be precision drilled for multi-positioning the three support pins to accomodate various size flat surfaces.

3.4.5 Electronic amplifier. The amplifier shall be transistorized; it shall be unit constructed of material currently used by the electronic gaging industry; and unless otherwise specified (see 6.2), it shall be operable from an input power supply of single phase 115 to 120 volts, AC, 60 Hertz. The amplifier shall have not less than two distinct measurement ranges with related read-out scales. Scale division values and measurement ranges shall be as specified in Table I. The measurement ranges may be direct, multi scale read-out with a scale selector or a scale changing control knob, or a single scale with a range selector control knob. The amplifier shall be equipped with an off-on switch, a calibration and zero setting selector, a built-in voltage regulator to compensate for line voltage fluctuation, a power supply and recorder plug-in socket and a carrying handle. Specific features not included herein shall be in accordance with Standard USASI C39.1-1964.

3.4.6 Range and scale divisions. Amplifiers with range and scale divisions exceeding those specified in Table I shall include the stated high range or smaller to the stated low range or larger.

TABLE I

AMPLIFIER SCALES	MEASUREMENT	
	RANGE	PER DIVISION
High	±0.00002	0.000,001
Low	±0.0002	0.000,010

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3.5 Electrical Equipment.

3.5.1 Power supply cord. The cord shall be a 6 foot minimum length, three wire oil resistant rubber insulated cord with voltage and current capacity not less than the comparator's amplifier rating. One wire shall serve as an effective ground to the amplifier and the opposite end of the cord shall terminate in a polarized three wire grounded plug that is applicable to the voltage specified. The power supply cord and its plug shall comply with requirements of UL 45.

3.6 Nameplate. A corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with Standard MIL-STD-130 shall be securely attached to each amplifier.

Nomenclature
Manufacturer's Model Number
Manufacturer's Serial Number
Special Characteristics, Volts, Watts, etc.
Stock Number
Manufacturer's Name
Contract Number
Date of Manufacture
US

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as quality conformance inspection.

4.2.1 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the gage assembly to pass the examination or one or more tests shall be cause for rejection. Quality conformance inspection shall consist of the following:

a. Examination (4.3).

b. Tests in (4.4).

c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.5).

4.3 Examination. The gage assembly shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

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4.4 Tests.

4.4.1 Operational Test. The gage assembly shall be operated for not less than 30 minutes. Proper operation of all controls, adjusting mechanisms and other accessories shall be inspected during the trial period.

4.4.2 Functional test. Amplifier meter response shall be not more than 1/4 second without overshoot. Zero point shall coincide on multi scale amplifiers. The gaging assembly shall be tested for repetitive accuracy in the following manner: The gaging head shall be positioned at three distinct locations on the column, the extreme low, the extreme high and center. Not less than three repetitive measurements shall be made at each location of the gage head and in each of the scale ranges. Measurement deviation, at any gage head location and all scale ranges, shall be not more than 1/10 scale division.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the requirements specified in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be levels A, B, or C in accordance with MIL-P-116, as specified (see 6.2).

5.1.1 Levels A & B. The requirements of MIL-P-116 shall be complied with in preservation and packaging using Method II for electronic amplifier and other electronic modules that will be involved, and Method I using P-6 preservative shall be used on mechanical components or modules.

5.1.2 Level C. For Level C packaging, comparator components shall be packaged in accordance with the manufacturer's standard commercial practice.

5.2 Packing. Packing shall be levels A, B, or C in accordance with MIL-P-116 as specified (see 6.2).

5.2.1 Level A. Comparator components preserved and packaged as indicated in 5.1.1 shall be overpacked into an exterior snug fitting, overseas, wood cleated plywood box, PPP-B-601, style optional. Exterior container insofar as practicable, shall be of a uniform size and minimum cube consistent with the protection required. Gross weight shall not exceed that as specified in the referenced container specification. Container shall be closed and sealed in accordance with the container specification and appendix thereto. Place package containing mechanical components or modules into bottom of container. Block and brace to prevent movement in accordance with MIL-STD-1186, using wood blocking and bracing. Cover with 1/4" plywood cut to interior dimensions (length and width) of container and secure in place as a divider using 2" X 2" wood cleats, nailed along both sides and ends (interior) of the container. The remaining packaged items consisting of the electronic amplifier and other electronic modules shall be packaged in the top section of the container. Fill all voids with cushioning material PPP-C-843.

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5.2.2 Level B. Comparator consisting of components preserved and packaged as indicated in 5.1.1 shall be packed the same as level A except that container shall be domestic type.

5.2.3 Level C. Comparator consisting of components, preserved and packaged as indicated in 5.1.1 or 5.1.2 shall be packed to assure carriers acceptance and safe delivery to destination at the lowest rates. Containers shall meet Uniform Freight Classification Rules or National Motor Freight Classification Rules, or regulations of other common carriers, as applicable to the mode of transportation.

5.3 Marking. In addition to any special markings as required by contract or order (see 6.2) markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The comparator will be used for determining ultra-precision dimensional values and accuracies of master inspection rings and discs, and for determining accuracies of other measuring devices.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Power supply requirements if different (see 3.2).

c. Height gaging range if different (see 3.4.1).

d. Contact tip if different (see 3.4.2).

e. Gaging anvil size if different (see 3.4.4).

f. In put power if different (see 3.4.5).

g. Level of preservation, packaging, and packing required (see 5.1 and 5.2).

h. Marking required (see 5.3).

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DOCUMENT IDENTIFIER AND TITLE

Electronically

Mil-C-83988(USAF) Comparator, Bench type, Vertical Column, Amplified

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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